

ASHUTOSH ROY

AI/ML Engineer

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SUMMARY

AI/ML Engineer with hands-on experience building end-to-end LLM, GenAI, RAG, and speech AI systems. Published researcher (RECCAP 2026, IIT Palakkad) with a co-authored paper from IIT Delhi. Skilled in fine-tuning transformer models, designing hybrid retrieval pipelines, deploying production ML systems via FastAPI, and building intelligent automation workflows. Experienced with NLP pipelines, multimodal AI systems, and scalable AI deployment using PyTorch, TensorFlow, LangChain, and MLOps-oriented workflows.

EXPERIENCE

Research Intern

Jan 2025 – Jun 2025

Indian Institute of Technology (IIT) Delhi

New Delhi, India

- Built an OCR-powered document understanding pipeline for archaeological datasets; benchmarked Tesseract, Marker, and Gemini OCR — selecting optimal strategy per document class to improve extraction accuracy.
- Designed hybrid retrieval (dense: FAISS/Pinecone + sparse: Apache Solr) for high-precision semantic + lexical document search over thousands of historical records.
- Deployed scalable **FastAPI** services for real-time LLM-based querying; integrated with frontend for sub-second retrieval responses.
- Co-authored **SARCH: Multimodal Search for Archaeological Archives** with IIT Delhi collaborators — submitted for peer review. arxiv.org/abs/2511.05667

AI/ML Engineer Intern

Oct 2024 – Nov 2024

MitoVoid AI

Remote

- Fine-tuned transformer-based NLP models (Mistral/LLaMA variants) for healthcare applications — medical record summarization and symptom-driven conversational AI.
- Reduced patient query resolution time by ~40% by deploying an AI chatbot for telemedicine with optimized intent understanding and response generation.
- Implemented GenAI-powered NLP workflows using LLMs and retrieval-augmented pipelines for intelligent healthcare automation and conversational AI systems.
- Built end-to-end ML pipelines: data preprocessing → model inference → API deployment for real-time healthcare applications.

PROJECTS

Speech Emotion & Stress Detection | PyTorch, Wav2Vec2, BiLSTM, Hugging Face | RECCAP 2026

Feb 2026 – May 2026

- Proposed a **self-supervised** Wav2Vec2 + BiLSTM multitask architecture for simultaneous 5-class emotion classification and continuous stress regression; achieved **85.9% accuracy, 0.856 F1, RMSE 0.1268**.
- Paper **accepted at RECCAP 2026, IIT Palakkad**; evaluated across 3 corpora (RAVDESS, TESS, SAVEE) with unified preprocessing (VAD, silence removal, log-Mel spectrograms).

Exam-Helper RAG System | Python, LLaMA 3, FAISS, Pinecone, LangChain, Streamlit

Dec 2024 – Apr 2025

- Built a production-ready LLM-powered contextual QA system on **LLaMA 3** with RAG; multimodal ingestion pipeline supports PDFs, images, and YouTube transcripts.
- Implemented **dense retrieval** via FAISS/Pinecone for sub-second similarity search, reducing hallucination and improving context precision.

Shopping Assistant Agent | Python, LangGraph, MCP, Antigravity, FastAPI

Sep 2024 – Dec 2024

- Engineered a multi-agent shopping assistant using **LangGraph** and MCP servers for intelligent product search, recommendation, and automation workflows with self-reflection and retry mechanisms

Medical Chatbot (LLM Fine-Tuning) | Python, Mistral 7B, QLoRA, Tesseract OCR

Mar 2024 – Apr 2024

- Fine-tuned **Mistral 7B** via **QLoRA** (4-bit quantization) for domain-specific medical NLU; reduced compute cost vs. full fine-tuning while preserving task accuracy.
- Built OCR pipeline using Tesseract to extract and structure information from unstructured clinical reports for downstream LLM inference.

TECHNICAL SKILLS

AI/ML & Research: LLMs, GenAI, Intelligent Automation, RAG, NLP, LangChain, Transformer Fine-Tuning (QLoRA/LoRA), Self-Supervised Learning, Multimodal AI, Speech Processing, Evaluation & Benchmarking, Dense Retrieval, Foundation Models

Libraries & Frameworks: PyTorch, Hugging Face, LangChain, TensorFlow, Scikit-learn, Pandas, NumPy, FastAPI, Streamlit, MLOps, Model Deployment

Databases & Search: FAISS, Pinecone, Apache Solr, MySQL

Languages: Python, C++, SQL

Tools: Git, Docker, Linux, Jupyter Notebook, Google Colab, VS Code, OpenAI GPT, Claude, Hugging Face Transformers

EDUCATION

Chhattisgarh Swami Vivekananda Technical University (CSVTU)

2022 – 2026

B.Tech (Hons) in Computer Science and Engineering (Artificial Intelligence)

Bhilai, Chhattisgarh

St. Columbus School

2020 – 2021

Higher Secondary (Class XII - Science PCMB) - 87%

Faridabad, Haryana